

Assignment 5 – MASH

A. Design a 2-2 CT-MASH based on 2 stages of 2nd-order 1-bit SDM. With 20kHz BW, define the lowest OSR to achieve peak SQNR of 100dB.

1). Show the FFT of 2-2 MASH final output. [2 point]

2). Assume that one of the reconstruction filters has a gain error of 1%, explain the impact of this non-ideality on the overall performance of the MASH. [1 point]

3). Add thermal noise at the 1st stage input to achieve 90dB SNR; Move the noise source to the input of 2nd stage input, with same SNR requirement. Compare two noise levels and explain the difference. [3 points]

4). Plot the output FFT for the case when the 2nd stage DAC is ideal, but the rise-time of the 1st stage DAC is equal to 20% of the clock period. Then plot the FFT for the case when the 1st stage DAC is ideal, but the 2nd stage DAC has a 20% rise time. Compare and explain the performance difference. [3 points]

5). Describe the difference in the design requirements of the 1st stage and 2nd stages. [1 point]